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Key Points:
- Enceladus' rapid spin requires higher-order theory for degree 2 gravity and shape
- Cassini gravity and shape imply a global ocean nearly but not completely frozen
- Ice shell much thinner over the South Polar Terrain facilitating plume venting

Supporting Information:
- Text S1 and Table S1

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Effect of Enceladus's rapid synchronous spin on interpretation of Cassini gravity

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Abstract Enceladus's degree 2 gravity, determined by Cassini, is nominally nonhydrostatic to 3σ (J_2/C_{22} = 3.38–3.63, as opposed to 10/3). less et al. (2014) interpret this in terms of a hydrostatic interior (core) and isostatic (not hydrostatic) floating ice shell. Enceladus's rapid (1.37 d) synchronous spin and tide distorts its shape substantially, though, enough that the predicted hydrostatic J_2/C_{22} is not 10/3 but closer to 3.25. This leads to the following revision to the internal picture of Enceladus, compared with less et al.: (1) the satellite's core is somewhat smaller and slightly denser (190 km radius and 2450 kg/m^3); (2) the compensation depth (shell thickness) of the global (degree 2) ice shell is ≈ 50 km, rather close to the base of the modeled ice + water layer; and (3) the compensation depth (shell thickness) beneath the South Polar Terrain (from J_3) remains shallower (thinner) at ≈ 30 km, independent of but influenced by the degree 2 solution.

1. Introduction

Enceladus, the diminutive but geologically hyperactive satellite of Saturn, is widely considered to be differentiated and possess an ocean, or at least a regional sea at the south pole, beneath its icy surface [*Spencer et al.*, 2009; *Spencer and Nimmo*, 2013]. Confirmation of the liquid water body, the presumed source of the numerous water-rich plumes erupting from the tiger stripe fissures near the south pole [*Porco et al.*, 2014], has come from the three passes of the Cassini Saturn orbiter devoted to measuring Enceladus' gravity field by means of Doppler tracking [*Iess et al.*, 2014]. The degree 2 potential (U_2) has been determined, and the only significant (≈ nonzero) terms are the zonal harmonic J_2 and the sectorial harmonic C_{22}, defined according to the standard definition in spherical coordinates,

$$U_2(r,\theta,\phi) = -\frac{GM}{r}\left(\frac{R}{r}\right)^2 [-J_2 P_{20}(\cos\theta) + C_{22}P_{22}(\cos\theta)\cos(2\phi)], \quad (1)$$

where R and M are the mean radius and mass of the satellite, G is the gravitational constant, and P_{20} and P_{22} are associated Legendre polynomials. The spherical coordinate system is chosen so that $\theta = 0$ is the (positive) spin axis and $\phi = 0$ is the sub-Saturn longitude. The measured values for J_2 and C_{22} are $(5435.2 \pm 34.9) \times 10^{-6}$ and $(1549.8 \pm 15.6) \times 10^{-6}$, respectively ($1\sigma$) [*Iess et al.*, 2014], smaller than the values predicted for a uniform density Enceladus [e.g., *McKinnon*, 2013, Table 1], and substantially so for J_2. This immediately implies that the outer layers of Enceladus are of lower density than the average, implying some degree of differentiation of rock from ice.

The degree 2 gravity does not directly lead to an estimate for Enceladus' mean moment-of-inertia (MOI) from the classic Radau-Darwin relation, as has been done for, say, the Galilean satellites [*Schubert et al.*, 2004]. As *Iess et al.* [2014] point out, the measured J_{22}/C_{22} ratio for Enceladus is 3.507 ± 0.042 (1σ), not the 10/3 expected for a synchronous satellite in hydrostatic equilibrium. *Iess et al.* [2014] creatively separate the gravitational potential into hydrostatic and nonhydrostatic parts, by assuming the interior (i.e., the rock core) is in hydrostatic equilibrium and the nonhydrostatic excess or anomalous surface topography is partially compensated. Enceladus' surface, in particular its zonal topography, does not follow its geoid, but it is clearly not completely uncompensated either. This is seen most strikingly in the J_2 gravity and in the negative zonal J_3 harmonic, the latter of which should have been best constrained on the two close south polar passes (E9 and E19).

The hydrostatic part of the solution implied an MOI of ≈ $0.335MR^2$, stated to be consistent with a low-density rock core (~2400 kg/m^3) overlain by a 60 km deep H_2O mantle (1000 kg/m^3) [*Iess et al.*, 2014]. Although

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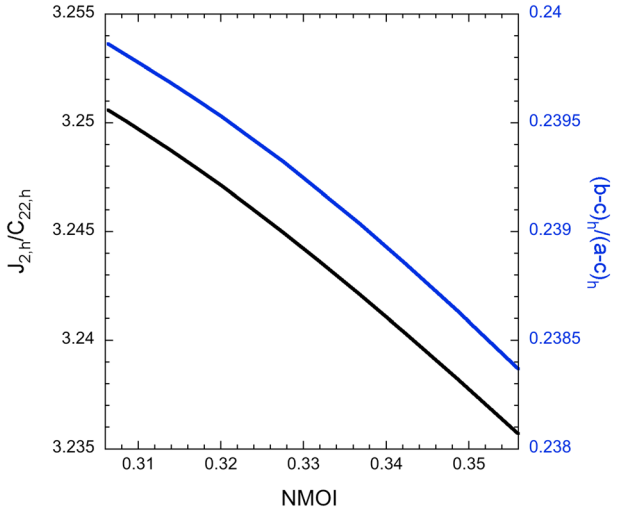


Figure 1. Equilibrium values of the J_2/C_{22} and $(b-c)/(a-c)$ ratios for Enceladus, as a function of its normalized mean moment-of-inertia (I/MR^2) calculated from a simple two-layer model (ice over rock).

different compensation mechanisms can be imagined, the most straightforward explanation for partial compensation of the gravity due to the excess topography is a floating ice shell of variable thickness (Airy isostasy). Classic Airy compensation then gives a compensation depth between 30 and 40 km, which can be thought of as the mean thickness of the ice shell. The consistent picture that emerges is that of a global ocean with a pronounced minimum shell thickness near the south pole, although the results have been portrayed as implying a south polar sea only (what provides the global compensation at degree 2 is unaddressed in such a portrayal). This paper reexamines the separation of the hydrostatic and nonhydrostatic components of the gravity solution using a higher-order theory of figures [e.g., *Tricarico*, 2014] and concludes that the hydrostatic component is smaller, the nonhydrostatic component larger, and that the degree 2 and degree 3 gravity are, within the uncertainties, sensitive to correlated but different aspects of the solution.

2. Effect of Rapid Rotation on Degree 2 Gravity and Shape

The degree 2 gravity harmonics J_2 and C_{22} and the axial ratios for hydrostatic, synchronous satellites increasingly deviate from classic first-order theory [e.g., *Hubbard*, 1984] with increasing spin and decreasing density (i.e., with increasing $q = \omega^2 R^3/GM$, where ω = rotation rate). The ratio $(b–c)/(a–c)$, where $a > b > c$ are the principal axes of the triaxial figure, is less than the classic 0.25. This is not a new result [*Chandrasekhar*, 1969, chapter 8; *Dermott and Thomas*, 1988], but it is underappreciated. In particular, to second order in q, J_2 increases less than linearly in q, whereas C_{22} increases greater than linearly (see *Tricarico* [2014] for a detailed discussion of recent developments in the theory of figures). The measured J_{22}/C_{22} ratio for Enceladus from *Iess et al*. [2014] might be better compared with a hydrostatic ratio of ~3.215 [*Tricarico*, 2014, Figure 6], not the canonical 10/3. In this case, Enceladus is actually nonhydrostatic at the 10% level. Now, the value of 3.215 strictly applies to a uniform density Enceladus, but the argument carries over to a differentiated body, discussed next.

Figure 1 illustrates the effect of Enceladus' rapid synchronous spin on its equilibrium triaxial figure and degree 2 gravity, as a function of its degree of differentiation. For simplicity, a two-layer model of the interior is assumed, a rock core of uniform density ρ_c, and an ice mantle of 925 kg m^{-3} (an ice mantle will be later argued to be a better approximation for Enceladus than a water mantle of 1000 kg m^{-3}). I utilize the recursion relations in Appendix G of *Tricarico* [2014], which are fourth order in the polar and equatorial eccentricities of both the surface and core, i.e., $e_p^2 = 1 - (a/c)^2$ and $e_{eq}^2 = 1 - (b/a)^2$. The solution is highly accurate for a moderately (≈3% at maximum) flattened body such as Enceladus. Core densities between 2000 and 3000 kg m^{-3} correspond approximately to normalized MOI values (NMOI) between 0.31 and 0.36. For this range, I find Enceladus's true equilibrium J_2/C_{22} ratio (or $J_{2,h}/C_{22,h}$) should lie between 3.235 and 3.25.

3. Separating Hydrostatic and Nonhydrostatic Degree 2 Gravity

Following *Iess et al*. [2014], the contribution of (possibly) compensated topography to the measured degree 2 gravity can be calculated as a function of compensation factor f_{20} ($= f_{22}$ as isotropic compensation at degree 2 is assumed), and subtracted from the total degree 2 gravity (either J_2 or C_{22}). Compensation factors indicate the extent to which the gravity due to *excess* surface topography is compensated at depth by a mass deficit or excess, and these factors vary by harmonic degree [e.g., *Wieczorek*, 2007]; e.g., $f = 0$ means there is no overall

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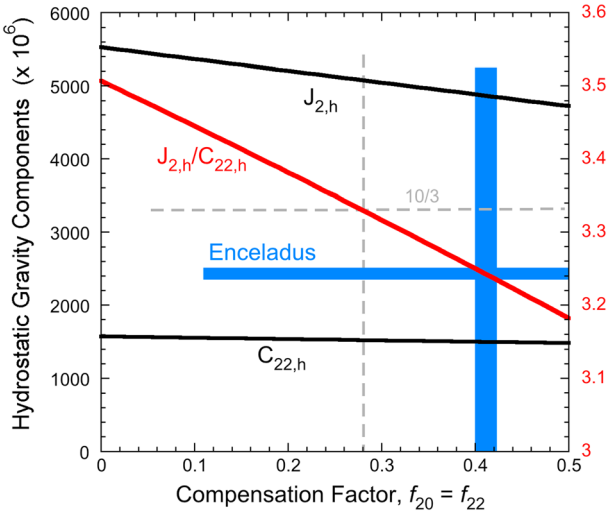


Figure 2. Enceladus degree 2 gravity as a function of topographic compensation factor. The left-hand axis indicates the Cassini measured values. Moving to the right, subtracting the gravitational contributions from excess topography, yields the hydrostatic contribution from Enceladus's internal structure (assuming the core *is* hydrostatic). The solution is self-consistent when the inferred hydrostatic values "$J_{2,h}$" and "$C_{22,h}$" reach the proper hydrostatic ratio [*Iess et al.*, 2014]. Note that the *Iess et al.* [2014] J_2 and C_{22} have been renormalized to $R = 252.1$ km from the $R = 254.2$ km used in the published gravity solution (L. Iess, personal communication, 2015) and that these renormalized values are used throughout this paper (Table S1).

contribution to the gravity signal, whereas $f = 1$ implies there is no compensation. The remainder is the *presumed* hydrostatic gravity, but even then only when $J_{2,h}/C_{22,h}$ is in the proper hydrostatic ratio (Figure 2). This ratio was assumed by *Iess et al.* [2014] to be 10/3, which gives an f at degree 2 of ≈ 0.27. For a differentiated, rapidly spinning Enceladus, however, this ratio is closer to 3.235–3.25, which from Figure 2 implies an $f_{20} = f_{22} = 0.40$–0.42. This compensation factor is not that small. Moreover, assuming Airy isostasy (shell thickness variation) and complete (i.e., iceberg-like) *physical* compensation implies a global depth of compensation d from $f_{20} = 1 - (1 - d/R)^2$ of ≈ 58 km.

For Airy isostasy, this compensation depth is the average thickness of Enceladus's global, floating ice shell. It is a global constraint determined from the measured values of J_2 and C_{22}, which were determined by the three flybys of varying geometry (E12 in the northern hemisphere and E9 and E19 in the southern; see Figure S2 in *Iess et al.* [2014]). The Airy interpretation implies a global ocean, albeit as will be shown below, one confined to the very bottom of the likely low-density (ice + water) layer. Physical isostasy implies an equator-to-pole shell thickness variation > 20 km; the azimuthal variation is smaller.

A fundamentally related, but more rigorous, approach estimates the admittances for the nonhydrostatic topography [*Hemingway et al.*, 2013b]. As detailed in the supplement to *Iess et al.* [2014], the degree 2 gravity field and topography can both be split into hydrostatic and nonhydrostatic components:

$$J_2 = J_{2,h} + J_{2,nh} \tag{2a}$$

$$C_{22} = C_{22,h} + C_{22,nh} \tag{2b}$$

$$H_{20} = H_{20,h} + H_{20,nh} \tag{3a}$$

$$H_{22} = H_{22,h} + H_{22,nh}, \tag{3b}$$

where H_{20} and H_{22} are the zonal and sectorial topographic spherical harmonic coefficients. The hydrostatic gravity *and* topography can be precisely calculated from the two-layer model above, and the remainder, the presumed nonhydrostatic components, are used to calculate the admittances from their ratios $Z_{20} = -J_{2,nh}/H_{20,nh}$ and $Z_{22} = C_{22,nh}/H_{22,nh}$. H_{20} and H_{22} are taken from *Nimmo et al.* [2011] (Table S1 in the supporting information).

These admittances are plotted in Figure 3 as a function of normalized MOI. The curves cross at an admittance of $\approx 4.8 \times 10^{-7}$ m^{-1} and a MOI $\approx 0.33MR^2$. By themselves, the admittances do not determine the source of the compensation at degree 2, but as long as the admittances are isotropic (a good assumption) and the rock core hydrostatic (more debatable, but possible), where the two admittances match determines the mean moment-of-inertia of the satellite. In terms of the two-layer model, an NMOI of 0.33 corresponds to a core density $\rho_c \approx 2450$ kg m^{-3} (see next section). Regarding uncertainties, the degree 2 gravity field is formally better determined (to ~1%) [*Iess et al.*, 2014] than the degree 2 topography (of order 5%) [*Nimmo et al.*, 2011; cf. *Thomas*, 2010]. Propagated errors near the crossover point in Figure 3 are ~15% for Z_{20} and ~30% for Z_{22} and are plotted. They imply that NMOI between ≈ 0.328 and ≈ 0.333 are acceptable gravity solutions. Note that Z_{22} switches sign for NMOI ≥ 0.336, which provides a quasi upper limit on the moment-of-inertia for Enceladus from this analysis method, given that negative admittances are rare (though see *Hemingway et al.*

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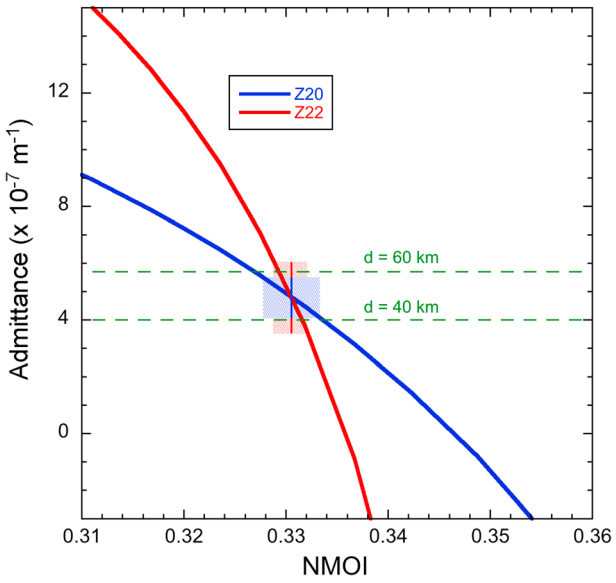


Figure 3. Admittance estimates (ratios of nonhydrostatic gravity to nonhydrostatic topography) determined from the difference between Cassini measured values [*Iess et al.*, 2014; *Nimmo et al.*, 2011] (Table S1) and estimates from the theory of figures (this paper), as a function of normalized moment-of-inertia. Admittance values at degree 2, Z_{20} (blue), and Z_{22} (red), should agree for zonally and sectorially isotropic topographic compensation; 1 sigma errors are shown. In this analysis, the entirety of the nonhydrostatic gravity signal is assigned to Enceladus's surface topography and an unspecified compensation mechanism. For comparison, admittance values for fully isostatically compensated, floating ice shells are shown (in green), as a function of compensation depth *d*.

[2013a] for possible examples on Titan); $Z_{22} = 0$ implies that the equatorial shape of Enceladus follows the local "geoid," a possibility noted in *McKinnon* [2013].

4. Internal Structure of Enceladus as a Function of Core Density and Moment-of-Inertia

Figure 4a plots core radius and density against mean NMOI for the two-layer structural model of Enceladus. The range in NMOI inferred from the admittance model above corresponds to core densities of 2375–2500 kg m^{-3} and mean core radii of 191–196 km. Compared with the values derived in *Iess et al.* [2014], the inferred ρ_c range is slightly higher but the core size and total shell thickness (≈60 km) are quite similar. Part of this similarity stems from choosing 925 kg m^{-3} for the ice + water shell (or "mantle") density in the present work, as opposed to the 1000 kg m^{-3} in Iess et al. In contrast, for a denser (or more water-rich) shell of 1000 kg m^{-3}, the same NMOI = 0.3305 ± 0.0025 would correspond to $\rho_c = 2530 \pm 70$ kg m^{-3} and a mean core radius = 185 ± 3 km (shell radius ~ 65 km). I note that the theory-of-figures solution (like that of lower-order Radau-Darwin) yields virtually the same J_2, C_{22}, and figure ellipticities for the same MOI.

Figure 4b shows the $J_{2,h}$ for the two-layer structural model of Enceladus used in the admittance calculation, again as a function of mean NMOI. For comparison, the range in $J_{2,h}$ determined from the simpler

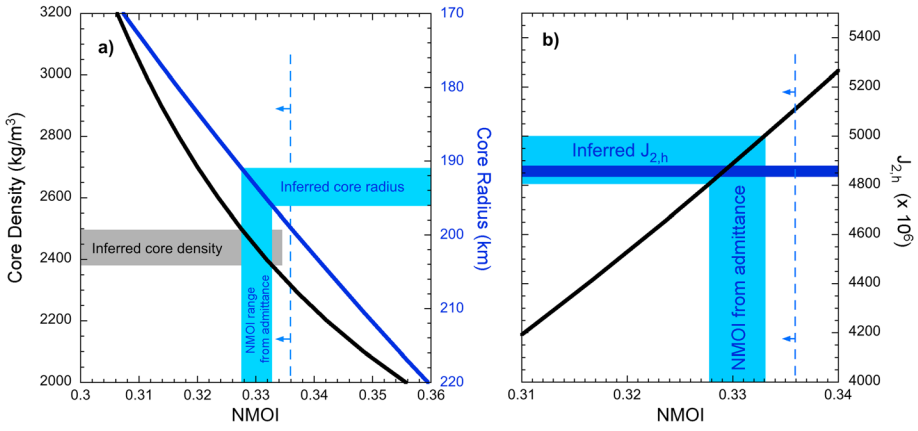


Figure 4. (a) Core density (black curve) and radius (blue curve) for a two-layer Enceladus model (ice shell of 925 kg m^{-3} over a rock core), as a function of NMOI. The NMOI inferred from the admittance calculation in Figure 3 (blue vertical bar) implies a core density close to 2450 kg m^{-3} and thus a core radius between 190 and 195 km; alternate structural interpretations are possible for the same NMOI range (see text). The dashed line indicates the possible upper limit on Enceladus's NMOI from $Z_{22} = 0$ admittance. (b) Hydrostatic zonal gravity ($J_{2,h}$) from the same model (reference radius = 252.1 km); the inferred range from the admittance solution overlaps the $J_{2,h}$ from the compensation model in Figure 2 (dark blue bar).

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compensation calculation in Figure 2 is plotted as the thin dark blue bar. The admittance solution and the simpler compensation calculation agree, but the admittance solution is more general and accurate—in that it utilizes more information from the theory of figures—and considers uncertainties, and so is preferred.

The admittance solution is agnostic as to the depth and source of the compensation. For Airy isostasy on a sphere as defined by *Phillips and Lambeck* [1980] and *Lambeck* [1988], however, the admittance for a floating, completely compensated shell can be written, following *Iess et al*. [2014], as

$$Z_{lm} = \frac{3\rho_s}{(2l+1)R\rho_E}\left[1-\left(1-d/R\right)^l\right] = \frac{3\rho_s}{(2l+1)R\rho_E} f_{lm} \quad (4)$$

where ρ_E is Enceladus's mean density, l and m are the spherical harmonic degree and order, and f_{lm} is the compensation factor referred to above. For a surface or shell density $\rho_s = 925\ \mathrm{kg\ m^{-3}}$ and an admittance value of $4.8\times10^{-7}\ \mathrm{m^{-1}}$ from Figure 3, a compensation depth of ≈ 49 km and a compensation factor of ≈ 0.35 is implied. Given the uncertainty in the admittance (perhaps $\pm 1\times10^{-7}\ \mathrm{m^{-1}}$) and that ρ_s in equation (4) strictly refers to the density of the nonhydrostatic surface topography (the topography whose mass excess or deficit must be compensated at depth) and neglects porosity, compensation depths between 40 and 60 km are acceptable. The admittances for these compensation depths are overlaid on Figure 3. I note that the porosity of Enceladus's impact battered, tectonically fractured, and/or plume fallback covered surface may be considerable, but this porosity should not extend through the entire ice shell [*Besserer et al*., 2013].

The compensation depth should be identified with mean depth of the ice shell. That it approaches the estimated total ice + water shell thickness (in the two-layer model) is significant. This implies the outer shell is largely but not completely frozen. This justifies the choice of $925\ \mathrm{kg\ m^{-3}}$ for the outer layer in the two-layer modeling and also shows than the Airy isostasy interpretation is at least self-consistent. For Airy isostasy, the topography at the base of the ice shell should reflect the nonhydrostatic surface topography, but multiplied by a factor of $(\rho_s/\Delta\rho)[R/(R-d)]^2$, where $\Delta\rho$ is the density difference at the shell/ocean interface. For $\rho_s = 925\ \mathrm{kg\ m^{-3}}$ and a $\Delta\rho = (1007-917) \approx 90\ \mathrm{kg\ m^{-3}}$ (referring to mildly saline water and ice at its melting point), and $d = 49$ km, this factor is 16. For the admittance match solution in Figure 3, $H_{20,\mathrm{nh}} \approx 1.3$ km, which implies a rather substantial equator-to-pole difference ≈ 30 km, weighed toward the poles. Specifically, the equatorial ice shell should be closer to ≈ 60 km thick (equatorially averaged) due to basal topography, while polar regions should be closer to ≈ 30 km thick. The implications of this variation are discussed below, but first it is important to fold in information from J_3.

5. Constraints From Zonal Degree 3 Gravity

The three Cassini passes ultimately yielded a measure of the degree 3 field or at least the antisymmetric zonal component J_3 [*Iess et al*., 2014] (Table S1). Measures or meaningful constraints on any other degree 3 terms were not possible, and the strength of the J_3 term is attributed to the "missing mass" associated with Enceladus's low-lying South Polar Terrain (SPT). Because there is no rotational or tidal component at degree 3, the admittance $Z_{30} = (3.1\pm0.6)\times10^{-7}\ \mathrm{m^{-1}}$ (1σ) can be immediately calculated from the degree 3 topography in *Nimmo et al*. [2011]. Z_{30} cannot be directly compared with Z_{20} or Z_{22}, however, except through a specific compensation model. For the case of isostatic Airy compensation (equation (4)), the degree 2 and degree 3 admittance results do not quite agree (see supporting information).

This mismatch may be more apparent than real, though. Determining higher-order gravity for such a small body as Enceladus is nontrivial, of course, and within 2σ the degree 2 admittance solution and Z_{30} are compatible. More to the point, the degree terms may be sensing "different parts of the elephant." That is, the admittance solution in Figure 3 is a global constraint determined from J_2 and C_{22}, which were determined by three flybys, over both the northern and southern hemispheres. In contrast, J_3 was probably most sensitive to the E9 and E19 tracks over the southern hemisphere, and whose closest approach points were at or near the south pole. *Iess et al*. [2014] estimate $f_{30} \approx 0.309$ (their Table S7), which yields a depth of compensation of 30 km. I would argue that is better thought of as the depth of compensation averaged over the South Polar Terrain, which need not be the same as the global depth of compensation. That is, the J_3 measurement is biased toward the thinness of the ice shell of the SPT.

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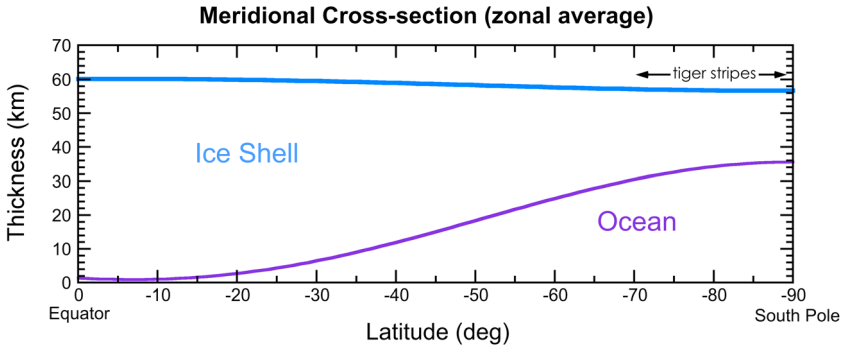


Figure 5. Cross section from Enceladus' equator to its south pole, for a representative admittance solution ($\rho_c = 2450$ kg m^{-3}; compensation depth $d = 48$ km at degree 2, 30 km at degree 3). The zonal average is shown, illustrating a nearly frozen shell at equatorial to low southern latitudes, whereas the shell over the actively venting portion of Enceladus (≈ 70–90° S) is ≤ 25 km thick. Along the equator, nonhydrostatic $lm = 22$ topography would thicken the shell by ~6 km at the sub- and anti-Saturn points, and thin it by the same amount at the leading and trailing points of motion. The former raises the possibility of grounding along the tidal axis, but this is argued in the text to be unlikely, or at least not extensive (and testable through libration measurements). A globally thick ice shell is more justifiable on energetic grounds [*Spencer and Nimmo*, 2013], and a thin basal ocean is a likely site for enhanced tidal heating [*Matsuyama*, 2014]. The cross section shown is based on an admittance model that assumes Enceladus possesses a hydrostatic rock core and that a floating (isostatic) ice shell of variable thickness is the source of Enceladus' nonhydrostatic gravity.

The inferred *additional* thinning at the SPT, with respect to the regional depth of compensation (that is, the polar shell thickness from Z_{20}), would be about 5 km, based on a similar isostatic argument as in the last section. This interpretation is consistent with the stereo-derived topography of the SPT. *Schenk and McKinnon* [2009] find the SPT topography to be remarkably flat overall, with a mean depth of ≈ 400–500 m measured *with respect to the best-fit ellipsoid* of *Thomas et al.* [2007]. For a regional average shell thickness of 30 km, this would imply an additional thinning of the southern polar ice by ≈ 6–7 km.

6. Discussion

6.1. Significance

The gravitational inverse problem is formally nonunique, as is well known. The forward model here (and in *Iess et al.* [2014]) satisfies Cassini gravity field constraints and is compatible with what we know about Enceladus. Figure 5 illustrates the shell and ocean thickness from equator to pole in the southern hemisphere, based on a representative admittance solution. Over the South Polar Terrain the ice shell is much thinner than at the equator, perhaps as little as 25 km overall. Measurements of degree 2 and degree 3 gravity are most accurate over the southern hemisphere. Taken at face value the shell in the admittance solution also thins toward the north pole, but there the degree 3 isostatic topography is positive and acts to limit the thinning. Nevertheless, it is a stretch to claim we have much information beyond degree 2 for the north.

What is most significant in the analysis presented here is the change in compensation depth at degree 2, which compared with the results in *Iess et al.* [2014], is almost but not quite the same as the total water + ice shell thickness (~50 versus ~60 km). The implication is that Enceladus possesses a global ocean, but that it is nearly frozen to its base. But the ocean cannot actually be frozen, with water confined to a regional sea at the south pole, or there would be no global Airy compensation [*Hemingway et al.*, 2013b]. Technically, the ice shell could be grounded at the equator (Figure 5), and especially at the sub- and anti-Saturnian points when the isostatic H_{22} topography is factored in, but this grounding cannot be so widespread as to compromise the compensation. The bottom topography is also critically dependent on the density of surface ice assumed in calculating its amplitude; even a modest (say, 20%) porosity in the upper few kilometers would have a proportional effect in reducing the amplitude of the ice "keel." Plus the core radius must be smaller when the average density of the ice + water shell is larger than 925 kg m^{-3}.

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6.2. Other Compensation Mechanisms

Given that the compensation depth is near the bottom of the ice + water shell (in the context to the two-layer ice over rock model interior), it is worth considering whether compensation could be provided by the core itself, as opposed to a water-ice interface. In such a situation, however, the anomalous core topography required would be opposite to that expected from dynamics [*McKinnon*, 2013], i.e., excess core mass would be needed at the poles and a deficit at the equator. This is unlikely. Alternatively, compensation in the shell could be provided in a Pratt sense, by a systematic shell density variation, but there is no obvious mechanism to make Enceladus's shell composition vary globally as a function of latitude and longitude (nor any independent evidence that this is so). Crustal porosity variations are possible in principle [*Schenk and McKinnon*, 2009; *Besserer et al*., 2013], but this would require the more heavily cratered north polar region of Enceladus to be less porous than the equatorial regions, which is counterintuitive. Moreover, the very shallow compensation depths appropriate to near-surface porosity would imply $f \approx 0$ [*Besserer et al*., 2013], inconsistent with Figure 2. I conclude that Airy isostasy due to a floating ice shell remains the logical choice as a compensation mechanism.

6.3. Future Work and Tests

Refined gravity models should take into account an improved topographic database from limb profiles and stereo imaging, as well as the explicit effects of (1) near-surface porosity, (2) the finite amplitude of basal shell topography [e.g., *Wieczorek*, 2007], and (3) nonhydrostatic core topography (if any). Independent evidence for a global, ungrounded ocean could come from measurement of an anomalously large libration amplitude (decoupled shell), which if measured could constrain the thickness of the floating ice shell.

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